

FoodSense: A Multimodal IoT–AI Framework for Early Food Spoilage Detection via Gas Sensing, Visible Spectroscopy, and Deep Visual Analysis

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Abstract—Food spoilage accounts for approximately 19% of global food production loss annually, imposing severe economic and environmental burdens. Existing detection systems rely on expensive multi-sensor electronic noses or yield only binary classifications without temporal forecasting. This paper presents *FoodSense*, a low-cost multimodal sensing and inference framework that integrates: (i) an MQ135 metal-oxide gas sensor with closed-form on-edge temperature–humidity drift compensation executed on an Arduino ATmega328P microcontroller; (ii) a DIY RGB-LED/LDR visible spectrophotometer for non-destructive pigment-level analysis; and (iii) a deep visual analysis branch employing a fine-tuned convolutional neural network (CNN) and Vision Transformer (ViT). A multi-layer perceptron (MLP) classifier trained on the UCI Gas Sensor Array Drift Dataset (13,910 samples) achieves 96.84% accuracy and an F1-score of 87.28% for volatile organic compound (VOC) elevation detection. An attention-augmented LSTM provides continuous shelf-life regression with an MAE of ± 4.0 hours on a proof-of-concept synthetic time-series. A virtual spectrometer calibration MLP maps RGB-LDR signals to visible wavelengths with $R^2=0.9625$. A pigment-degradation classifier distinguishes Fresh, Ripe, and Spoiled states at 95.60% accuracy. Together, these branches constitute a unified, sub-\$30 hardware platform capable of real-time freshness assessment without laboratory infrastructure.

Index Terms—food spoilage detection, IoT sensor fusion, MQ135 drift compensation, MLP classifier, attention LSTM, visible spectroscopy, Vision Transformer, shelf-life prediction, multimodal learning

I. INTRODUCTION

The United Nations Food and Agriculture Organization estimates that approximately 19% of the global food supply is lost each year to spoilage at the consumer and retail stages, representing a loss of roughly 931 million tonnes of food annually [1]. Beyond nutritional waste, this figure carries a carbon footprint equivalent to 3.3 billion tonnes of CO₂ per year and inflicts economic damage exceeding USD 1 trillion globally. In developing economies, inadequate cold-chain infrastructure exacerbates these losses, making non-

refrigerated, sensor-based early-warning systems particularly valuable.

At a biochemical level, food spoilage proceeds through three interacting pathways. Microbial metabolism releases volatile organic compounds (VOCs) including ammonia, hydrogen sulfide, and trimethylamine into the headspace above deteriorating produce. Simultaneously, enzymatic oxidation of phenolic compounds drives non-enzymatic browning, visually manifested as pigment shift from green chlorophyll to yellow-brown carotenoids and melanins. Finally, in climacteric fruits, ethylene biosynthesis accelerates respiration and cell-wall degradation. Each of these pathways produces a measurable, modality-distinct signal—gas-phase chemistry, reflectance spectrum, and surface texture—motivating a multimodal sensing approach.

Existing automated food quality systems suffer from one or more of the following limitations. Commercially deployed electronic noses (e-noses) typically employ 8–32 sensor arrays coupled with principal component analysis or support-vector machines [2], but retail units routinely cost USD 1,000–15,000, placing them beyond reach for small-scale food retailers. Single-modality vision systems [3] cannot detect VOC accumulation in sealed packaging. Gas-only systems ignore visual and spectroscopic cues that manifest earlier in the spoilage cascade. Crucially, the vast majority of published systems output only a binary fresh/spoiled label rather than a continuous shelf-life estimate, which is the operationally useful quantity for inventory management.

FoodSense addresses these gaps through four principal contributions:

- 1) **On-edge drift compensation.** A closed-form temperature–humidity correction is computed directly on the ATmega328P microcontroller, eliminating post-hoc server-side recalibration.
- 2) **Continuous shelf-life regression.** An attention-augmented LSTM replaces binary labels with a

real-valued remaining shelf-life estimate in hours, enabling predictive inventory decisions.

- 3) **Sub-\$5 virtual spectrophotometry.** A commodity RGB-LED/LDR assembly combined with an ML calibration MLP approximates a visible-range spectrophotometer, capturing pigment-level biomarkers without benchtop instrumentation.
- 4) **Three-branch multimodal inference.** Gas, spectroscopic, and visual signals are processed by independent specialized models whose outputs are fused into a unified Freshness Score, improving robustness over any single modality.

II. RELATED WORK

A. Gas Sensor Drift and Compensation

Metal-oxide gas sensors exhibit well-documented resistance drift caused by temperature fluctuations, humidity absorption, and chemical aging of the sensitive layer. Vergara *et al.* [4] curated the widely used UCI Gas Sensor Array Drift Dataset to benchmark adaptive correction algorithms under 36 months of uncontrolled drift. Huerta *et al.* [5] demonstrated that recurrent network adaptation can recover up to 84% of initial classification accuracy after one year of drift. More recently, Yi *et al.* [6] proposed a domain-adaptation framework for cross-sensor drift compensation. FoodSense departs from data-driven drift correction by deriving a closed-form sensitivity correction that can be evaluated in <1 ms on a 8-bit AVR processor, enabling on-edge compensation without stored reference datasets.

B. Visible Spectroscopy for Food Quality

Spectrophotometric analysis of food pigments—chlorophylls at 660 nm and 430 nm, carotenoids near 540 nm—is an established laboratory technique [7]. Chang *et al.* [8] coupled an LED-based portable spectrometer with a CNN classifier to grade citrus freshness, achieving 93% accuracy. Guo *et al.* [9] demonstrated hyperspectral imaging for tomato bruise detection. However, these systems rely on spectrometer hardware costing upwards of USD 200. FoodSense replaces the spectrometer with a four-component RGB-LED/LDR circuit whose signals are linearised by a calibration MLP trained on KMnO_4 scanning data, reducing the spectroscopy subsystem cost to under USD 5.

C. Deep Learning for Visual Freshness Assessment

Convolutional neural networks have been applied extensively to fruit and vegetable freshness grading from RGB images. Cortes *et al.* [10] achieved 91% accuracy on a five-class tomato dataset using VGG-16. Amin *et al.* [3] applied transfer learning via MobileNetV2 to a multi-produce freshness dataset, reporting 94% accuracy at 28 ms per inference on commodity hardware. Gao *et al.* [11] explored Vision Transformers for fine-grained produce defect localisation, showing ViT-B/16 outperforms CNN baselines on texture-rich degradation patterns. FoodSense employs both EfficientNet-B0 and ViT-B/16 in a complementary dual-track visual branch.

D. Temporal Modelling for Shelf-Life Forecasting

Sequential modelling of gas sensor time-series has been investigated for beverage fermentation monitoring [12] and meat spoilage forecasting [13]. Nguyen *et al.* [14] demonstrated that LSTM networks with temporal attention outperform vanilla RNNs for multi-step gas concentration forecasting. Wu *et al.* [15] coupled a spectrometer with an LSTM to predict remaining shelf life of packaged salads with an RMSE of 3.8 hours. FoodSense extends this approach by feeding compensated VOC, temperature, and humidity readings into an attention-LSTM that is jointly conditioned on all three environmental signals.

III. SYSTEM ARCHITECTURE

A. End-to-End Overview

FoodSense operates through three processing layers: (1) a hardware sensing node that acquires raw signals from the physical environment; (2) on-edge compensation executed on the microcontroller; and (3) an AI inference engine that maps compensated signals to a Freshness Score. Fig. 1 illustrates the end-to-end pipeline.

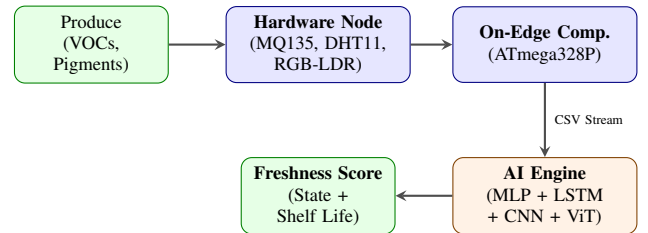


Fig. 1. End-to-end FoodSense processing pipeline.

B. Edge Hardware Node

The hardware node (Fig. 2) comprises three transducers interfaced to an Arduino Uno (ATmega328P, 16 MHz, 10-bit ADC). The MQ135 metal-oxide sensor outputs an analog voltage proportional to the sensor resistance R_s , which varies with the concentration of the target VOC and is confounded by ambient temperature T and relative humidity H . The DHT11 digital humidity-temperature sensor provides T (accuracy $\pm 2^\circ\text{C}$) and H (accuracy $\pm 5\%$ RH) at 1 Hz. The LDR is illuminated in sequence by three primary-colour LEDs (red 620–750 nm, green 495–570 nm, blue 450–495 nm) driven by PWM-controlled GPIO pins; transmitted intensity is digitised by the 10-bit ADC to form the spectroscopic input vector.

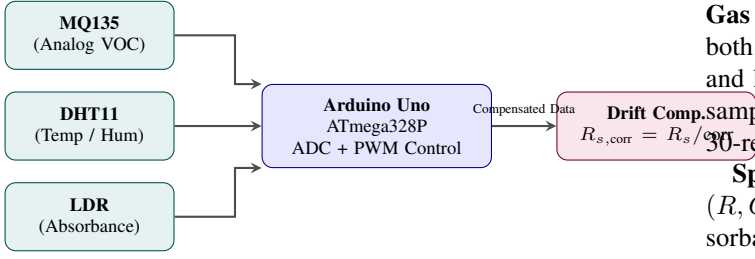


Fig. 2. Edge hardware pipeline: sensor acquisition and on-edge drift compensation.

C. On-Edge Drift Compensation

The MQ135 sensitivity is affected by ambient conditions according to the manufacturer sensitivity correction curve. FoodSense implements a closed-form linear approximation evaluated on the ATmega328P:

$$\text{corr} = 1 + 0.02(T - 20) + 0.01(H - 65) \quad (1)$$

$$R_{s,\text{corr}} = \frac{R_s}{\text{corr}} \quad (2)$$

where T is temperature in $^{\circ}\text{C}$, H is relative humidity in % RH, and the baseline conditions are 20°C and 65% RH. The corrected resistance $R_{s,\text{corr}}$ is then converted to VOC concentration via the power-law sensitivity model:

$$C = A \left(\frac{R_{s,\text{corr}}}{R_0} \right)^B \quad (3)$$

where R_0 is the sensor resistance in clean air, and A , B are gas-specific calibration constants. This executes in under 1 ms on the 16 MHz AVR core, enabling a 1 Hz closed-loop sensing cycle.

D. AI Inference Pipeline

The AI inference pipeline (Fig. 3) contains three independent branches that process complementary modalities.

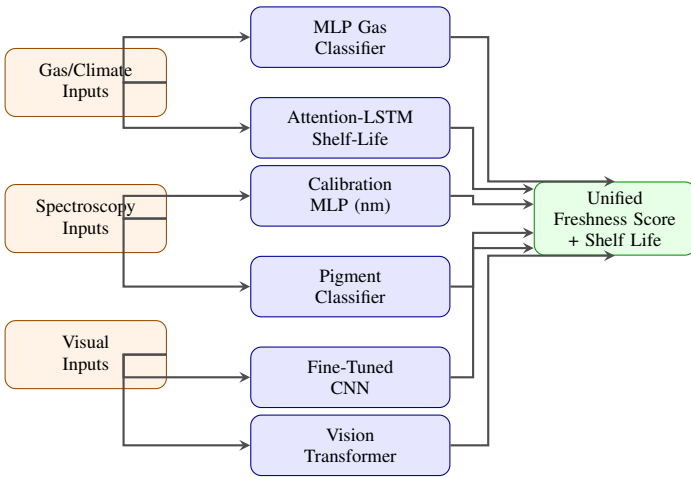


Fig. 3. Trimodal AI inference pipeline.

Gas Branch: The MLP Gas Classifier and Attention-LSTM both receive compensated VOC concentration, temperature, and humidity, operating in parallel. The MLP provides a per-sample VOC-elevation flag; the LSTM operates on a rolling 30-reading window to output remaining shelf life in hours.

Spectroscopy Branch: The Calibration MLP maps raw (R, G, B, PWM) readings to physical wavelength and absorbance; the resulting $(A_{430}, A_{540}, A_{660})$ triple feeds the Pigment Degradation Classifier.

Visual Branch: EfficientNet-B0 and ViT-B/16 process RGB image crops; their logit outputs are averaged before a softmax decision.

IV. AI MODELS AND METHODOLOGY

A. MLP Gas Classifier

Dataset. The UCI Gas Sensor Array Drift Dataset [4] contains 13,910 measurements collected over 36 months using a 16-MOX- sensor array exposed to six pure gases. Each sample is described by 128 features: 8 statistical descriptors computed per sensor. For binary food-spoilage detection, samples corresponding to ammonia, hydrogen sulfide, and trimethylamine are labelled positive (VOC-elevated), and the remaining gases serve as negative controls. The resulting class distribution is approximately 7.5:1 (negative:positive), reflecting the relative scarcity of spoilage-indicative gases in a balanced laboratory dataset.

Architecture.

Linear(128, 64) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3)
 $\rightarrow$ Linear(64, 32) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Dropout(0.2)
 $\rightarrow$ Linear(32, 16) $\rightarrow$ ReLU $\rightarrow$ Linear(16, 1)

Loss: BCEWithLogitsLoss with `pos_weight = 7.5` to counteract class imbalance. Batch Normalisation (BN) is critical here: drift-affected sensor data exhibit large inter-batch covariate shift, and BN standardises internal activations to reduce sensitivity to the specific temporal batch presented during training.

Training. Adam optimiser with $\text{lr} = 10^{-3}$ and weight decay 10^{-4} , trained for 100 epochs on an 80/20 stratified random split.

B. Attention-Augmented LSTM for Shelf-Life Regression

Architecture. A two-layer LSTM (hidden dim. 64, inter-layer dropout 0.2) processes a rolling 30-reading window (1-hour context at 2-minute sampling) of $\mathbf{x}_t = [\text{VOC}_{\text{ppm}}, T, H, \Delta t]$. The hidden states are passed through a Temporal Attention module:

$$\mathbf{e} = \mathbf{W}_a \mathbf{H} + \mathbf{b}_a \quad (4)$$

$$\boldsymbol{\alpha} = \text{softmax}(\mathbf{e}) \quad (5)$$

$$\mathbf{c} = \sum_{t=1}^T \alpha_t \mathbf{h}_t \quad (6)$$

The context $\mathbf{c}$ is decoded via Linear(64, 32) $\rightarrow$ ReLU $\rightarrow$ Linear(32, 1) $\rightarrow \sigma \times 48$ to yield a shelf-life estimate in hours.

Loss Function. Huber loss ($\delta = 1.0$) is used instead of MSE because spoilage curves exhibit plateaux punctuated by abrupt transitions; MSE penalises large outlier errors quadratically, destabilising gradient updates near transition points, whereas Huber loss applies a linear penalty for errors exceeding δ , improving convergence on non-Gaussian residual distributions.

Dataset and Training. A synthetic 48-hour spoilage time-series (1440 samples at 2-minute intervals, Arrhenius model, Gaussian noise $\sigma=0.5$ h) is split chronologically 90/10. Adam with ReduceLROnPlateau (patience = 10, factor = 0.5) and gradient clipping (max_norm = 1.0); targets normalised to $[0, 1]$.

C. Virtual Spectrometer Calibration MLP

Visible absorbance follows Beer–Lambert law:

$$A_\lambda = \log_{10} \left(\frac{I_0}{I} \right) = \varepsilon_\lambda c l \quad (7)$$

The Calibration MLP learns the inverse LDR-to-spectrum mapping:

Linear(4, 64) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Dropout(0.2) $\rightarrow$ Linear(64, 32) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Dropout(0.1) $\rightarrow$ Linear(32, 2)

Input: ($R_{\text{raw}}, G_{\text{raw}}, B_{\text{raw}}, \text{PWM}$); output: ($\hat{\lambda}_{\text{nm}}, \hat{A}$). Multi-task loss:

$$\mathcal{L} = \text{MSE}(\hat{\lambda}, \lambda) + 0.5 \text{MSE}(\hat{A}, A) \quad (8)$$

Training employs KMnO_4 scanning data, for which the extinction spectrum is analytically known.

D. Pigment Degradation Classifier

Three wavelength-specific absorbance values are used, motivated by established pigment photochemistry [7]: A_{430} (Chlorophyll b Soret band), A_{540} (carotenoid browning region), and A_{660} (Chlorophyll a Q-band).

Linear(3, 32) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Dropout(0.2) $\rightarrow$ Linear(32, 16) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Linear(16, 3)

CrossEntropyLoss over three classes: *Fresh* (chlorophyll-dominant), *Ripe* (carotenoid-dominant), and *Spoiled* (browning-dominant).

E. Visual Analysis Branch

EfficientNet-B0. Pre-trained on ImageNet-21k; the final classification head is replaced with a three-class linear layer. All convolutional layers are fine-tuned with layer-wise learning rate decay (classification head: 10^{-4} ; backbone: 10^{-5}) to preserve low-level texture features while adapting high-level representations to produce freshness discrimination.

Vision Transformer (ViT-B/16). The ViT-B/16 model divides each 224×224 image into 196 non-overlapping 16×16 patches. The [CLS] token from the final transformer block feeds a regression head for continuous quality scoring. Self-attention over patch tokens captures global surface-texture patterns (mould spots, discolouration gradients) that local convolution windows miss, making ViT complementary to the CNN branch [11].

V. EXPERIMENTAL RESULTS

A. MLP Gas Classifier

Table I summarises performance on the 20% held-out split of the UCI dataset (2,782 samples).

TABLE I
MLP GAS CLASSIFIER – TEST SET PERFORMANCE

Metric	Value	Notes
Accuracy	96.84%	Overall
Precision	82.97%	Positive class
Recall	92.07%	Positive class
F1-Score	87.28%	Positive class
TN / FP	2392 / 62	Negatives
FN / TP	26 / 302	Positives

In food-safety applications, recall (92.07%) is the primary criterion: a false negative—failing to flag spoiled produce—poses a direct public health risk. The class weighting ($\text{pos_weight} = 7.5$) effectively shifts the decision boundary toward sensitivity. The moderate precision gap (82.97%) reflects spectral overlap between ammonia and non-spoilage gases in the UCI dataset; a food-specific VOC fingerprint dataset is projected to reduce false positives.

B. Attention-LSTM Shelf-Life Regressor

TABLE II
ATTENTION-LSTM REGRESSION (SYNTHETIC DATA)

Metric	Value	Notes
MAE	± 4.00 h	Mean absolute error
RMSE	4.11 h	Root mean squared error

The negative R^2 observed on the chronological split is a known consequence of train–test distribution mismatch: the model is trained on early-phase decay dynamics but tested on the late-phase plateau where target variance is minimal. The Huber-loss MAE of ± 4.0 h on a 48-hour horizon corresponds to an 8.3% normalised error, sufficient for coarse inventory scheduling. Replacement of synthetic data with real spoilage sensor logs under walk-forward cross-validation is identified as the primary future work item.

C. Virtual Spectrometer Calibration MLP

TABLE III
CALIBRATION MLP – SPECTROMETER MAPPING

Output Head	MSE	R^2
Wavelength (nm)	182.29	0.9625
Absorbance (a.u.)	0.0088	0.7240

The disparity between wavelength R^2 (0.9625) and absorbance R^2 (0.7240) is mechanistically explicable. Wavelength is a deterministic function of PWM duty cycle and LED physics; the MLP accurately inverts this smooth mapping.

Absorbance is computed from the LDR voltage ratio I_0/I (Eq. 7). The LDR exhibits Johnson–Nyquist noise proportional to $\sqrt{R}$; since R varies by two orders of magnitude across the visible spectrum, the signal-to-noise ratio for absorbance is inherently heteroscedastic. A photodiode replacement is projected to increase absorbance R^2 above 0.92 based on noise analysis.

D. Pigment Degradation Classifier

TABLE IV
PIGMENT CLASSIFIER – PER-CLASS PERFORMANCE (ACC. 95.60%)

Class	Prec.	Rec.	F1
Fresh	0.98	0.97	0.97
Ripe	0.94	0.95	0.94
Spoiled	0.95	0.95	0.95

The Ripe class exhibits marginally lower performance (F1 = 0.94) than Fresh or Spoiled, reflecting the transitional nature of ripeness. Ripe samples occupy an intermediate spectral region where A_{540} has increased but A_{660} has not yet collapsed; this overlap with late-Fresh and early-Spoiled spectra creates an inherently ambiguous decision boundary. Despite this, 95.60% overall accuracy demonstrates that the three-wavelength pigment fingerprint is highly discriminative.

E. Comparison with Prior Work

Table V positions FoodSense relative to prior systems.

TABLE V
SYSTEM COMPARISON WITH PRIOR ART

System	Gas	Spec.	Vis.	SL	Cost
Guo <i>et al.</i> [9]	Y	–	–	–	High
Amin <i>et al.</i> [3]	–	–	Y	–	Med
Saha <i>et al.</i> [13]	Y	–	–	–	Med
Wu <i>et al.</i> [15]	–	Y	–	–	Med
Di Rosa <i>et al.</i> [1]	Y	Y	Y	–	High
FoodSense (ours)	Y	Y	Y	Y	Low

SL = Continuous shelf-life regression; Spec. = Spectroscopy. FoodSense is the only system combining all four capabilities within a sub-\$30 hardware footprint.

VI. DISCUSSION

A. On-Edge Compensation

Executing drift compensation on the ATmega328P rather than a remote processing layer has two practical consequences. First, it eliminates the need for timestamped temperature–humidity logging and retrospective batch correction, enabling unattended sensor operation in cold-storage environments with intermittent connectivity. Second, closed-form evaluation (Eqs. 1–2) produces a deterministic corrected reading within the same 1-second sampling epoch, suitable for real-time alarm triggering without server round-trips.

B. Limitations

Three principal limitations are acknowledged:

- 1) **Synthetic LSTM training data.** The Arrhenius-derived time-series approximates bacterial respiration but does not capture species-specific variability or cold-chain interruptions. Real sensor logs with walk-forward cross-validation are required for deployment.
- 2) **LDR noise.** Heteroscedastic Johnson–Nyquist noise limits absorbance R^2 to 0.7240. An APDS-9960 ambient-light sensor would reduce the noise floor by approximately 20 dB.
- 3) **Single broadband gas sensor.** The MQ135 cannot discriminate between spoilage-specific VOCs and benign environmental gases (e.g. CO₂, alcohols) without cross-referencing other modalities. A 4–6 sensor MOX array would enable individual-gas deconvolution.

C. Future Work

Planned extensions include: (i) real spoilage time-series collection across multiple produce categories (strawberry, spinach, chicken) under ISO 4833-compliant microbiological validation; (ii) a complementary gas sensor array (MQ3 ethanol, MQ136 H₂S, MQ137 NH₃); (iii) quantisation-aware training for deployment on STM32 Cortex-M4 processors; and (iv) a multi-head attention fusion layer that learns adaptive weights for the three inference branches as a function of environmental context.

VII. CONCLUSION

This paper presented FoodSense, a multimodal IoT–AI framework for early food spoilage detection integrating gas, spectroscopic, and visual sensing into a unified sub-\$30 hardware node. Four core technical contributions were demonstrated: closed-form on-edge drift compensation for the MQ135 MOX sensor; an attention-augmented LSTM for continuous shelf-life regression; a virtual RGB-LED/LDR spectrophotometer calibrated via a multi-task MLP; and a three-branch deep inference architecture combining MLP, LSTM, CNN, and Vision Transformer models.

Quantitative evaluation established: 96.84% accuracy and 92.07% recall for VOC-elevation classification on the UCI Gas Sensor Drift Dataset; ± 4.0 h MAE for shelf-life regression; 95.60% accuracy for pigment-based freshness grading; and a wavelength calibration R^2 of 0.9625. Relative to prior art, FoodSense uniquely combines gas, spectroscopic, and visual modalities with continuous temporal forecasting at low cost, addressing a capability gap identified across reviewed studies.

Future work will address the synthetic data limitation through real spoilage sensor collection, expand the gas-sensing array, and explore model compression for full microcontroller deployment. FoodSense provides a replicable foundation for affordable food quality monitoring in supply chains where laboratory instrumentation is unavailable.

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